

Ideas Parked — Not Yet Merged (v17)

Staging document; each entry marks its eventual destination

A full resolution to Gap 2 of ideas_v1 — “Location as a sort,” the open item that survived untouched through sixteen staging versions while Gap 1 was closed. Worked out and verified over several turns; recorded here as one connected entry rather than scattered across the conversation that produced it.

Blocking condition, stated before anything else in this file: this resolution cannot be merged into Paper 1 on its own. It depends on the meta-vector taxonomy itself — information vectors versus meta vectors, as a formal distinction sitting on top of Axiom 7 — being imported first, as its own prior step. That taxonomy is already destined for Paper 1 (ideas_v11), but is not yet merged there. Everything below is a specification of what goes *into* that taxonomy once it exists, not a thing that can be written into the current paper by itself.

1 Gap 2, restated from source

ideas_v1’s own words, unedited: *“Location as a sort — a third candidate. The existing paragraph names two candidates and adopts neither. A third was tested here: dedicate separate, disjoint dimensions to content, to time, and to location, rather than letting any of them share an axis.”* Verified there: two shifts through disjoint dimensions commute exactly; content stays recoverable after both. Left open there, in its own words: *“The encoding itself — what specific rotation a given time value or location corresponds to — is not stated anywhere. The geometry now supports it cleanly; the mapping does not exist yet.”* That is the specific gap this entry closes.

2 The architectural decision: meta vector, not sort

Tested against the paper’s own existing distinction, not decided by preference. The cap $r \leq 1$ encodes strength of *content* — how activated a claim is. Location and time are not gradable claims; they are coordinates describing where or when content applies. Checked against the current paper directly: this is the same line already drawn for Assembly’s own seat, word for word — *“A seat is external bookkeeping — a place a unit occupies, carrying no meaning and entering no chart. A location sort would be a unit in its own right...and would have to pass Axiom 0 on its own terms.”* Meta vector is the seat-shaped answer; sort is the full-unit-shaped one. Time has its own independent precedent pointing the same way: the existing time-bucket metadata is already described as *“a fact about how a particular walk was assembled, not a property its occupant carries into the next one”* — meta in spirit, already, before this entry names it so formally.

The cost, named rather than hidden: a sort gets Mix, Bundle, and T for free once admitted; a meta vector does not, and needs its own grouping mechanism instead. That mechanism is already sketched, separately, destined for Inference (ideas_v11): querying several atoms’ meta vectors, confirming they sit close, promoting one shared value up to the group.

3 Location: one vector, verified

Standard (θ, ϕ) from latitude and longitude, the same structure $\hat{n}$ already has, with zero adaptation needed: latitude is bounded $[-90^\circ, 90^\circ]$, matching θ 's own bounded range exactly; longitude is cyclic $[-180^\circ, 180^\circ]$, matching ϕ exactly. Verified directly on real coordinates — London ($51.5^\circ\text{N}, 0.1^\circ\text{W}$) $\rightarrow (0.6225, -0.0011, 0.7826)$, Tokyo ($35.7^\circ\text{N}, 139.7^\circ\text{E}$) $\rightarrow (-0.6194, 0.5252, 0.5835)$, Sydney ($33.9^\circ\text{S}, 151.2^\circ\text{E}$) $\rightarrow (-0.7273, 0.3999, -0.5577)$ — all unit norm to six decimal places, all distinct. Ready as stated; no open question remains here.

4 Time: the obvious encoding fails, verified precisely; the fix does not

The naive mapping (hour on θ , minute on ϕ) was tested, not assumed, and broke in two distinct ways. First, a pole collapse: at hour = 0, every minute value lands at the identical point $(0, 0, 1)$, since $\theta = 0$ is a pole and every ϕ is degenerate there. Second, and worse: 11:59 and 0:00, one minute apart, land at Euclidean distance ≈ 1.99 — antipodal, the maximum possible separation on the sphere — because θ was never built to wrap, and sweeping hour across $[0, \pi]$ treats the end of the cycle as the opposite pole from its start rather than its neighbor.

The fix stays inside existing machinery and was verified to work. Put minute on ϕ (already cyclic) and hour on χ (the atom's own existing phase coordinate, also already cyclic), holding θ fixed at the equator so neither hand ever touches a pole. Verified: 11:59 and 0:00 now land 6° apart in minute and under 1° apart in hour, both hands fully independent. θ , freed from a job it was never suited for, sits unused — available for a third hand (seconds) if ever wanted, which would make this a structurally exact match for a standard three-hand analog clock: two coordinates already on the atom, one still spare.

5 Timestamp: a register, not a scalar, scaled to N

Not one timestamp per atom — one per contributing vector. A simple atom ($N = 1$) carries one; a Collapsed atom carries N , since different contributions can genuinely have been born at different moments, and Collapse's whole design principle is keeping every contribution rather than averaging it away. This costs nothing new to keep synchronized: UnCollapse already removes exactly one contribution, $N \rightarrow N - 1$ (checked directly against the current paper's own stated rule), so the same operation removes that contribution's timestamp alongside it, automatically.

6 Age: derived, never stored, and cannot correctly be otherwise

age = now — timestamp, computed on demand, the same shape as V_{derived} and Axiom 7's seven metrics — all read from stored data, none independently stored themselves. This is not a convenience; a stored age would be wrong the instant time passed, so deriving it is the only version that is ever actually correct. “Now” itself is external context the formal system assumes rather than defines — the same status the paper already gives lexicon standardization: *“sociological and empirical; it is not an axiom.”*

7 Expiry: a second, genuinely new register — not derivable, unlike age

Raised one turn after timestamp, under the same phrase (“time period”) age had already used — flagged and separated here so the phrase does not carry two meanings going forward: **age** is how old, derived; **expiry** is when a contribution stops being valid, and unlike age, it cannot be computed from timestamp alone. No universal rule maps birth-time to shelf-life — different contributions to one Collapsed atom plausibly expire at different rates (three sensor readings — humidity, pressure, windspeed — collapsed together need not go stale together). Expiry is therefore genuinely new information, requiring its own register, same length as the timestamp register, shrinking in lockstep with it under the same $N \rightarrow N - 1$ UnCollapse rule. It yields its own derived check, the same shape as age: $\text{now} < \text{expiry}_i$, computed per vector, never stored.

Left deliberately open. Whether Mix or Collapse should automatically skip expired contributions, or whether validity filtering is a separate reading applied before either operator runs, is a real design decision and is not answered here.

8 Two caveats, kept honest rather than smoothed over

Timestamps help disambiguate the degenerate UnCollapse case (two contributions sharing identical $(r, \hat{n}, \chi)$) most of the time, but are not a guaranteed substitute for the dedicated *id* field from ideas_v9: two contributions ingested in the same batch could share both content and birth time, and only a construction-guaranteed *id* — not a timestamp — would still separate them.

9 Inventory, and where each piece actually lands

Five meta-vector purposes now specified: V_{derived} and link vectors (already named, ideas_v11), location, the timestamp register, and the expiry register. Per ideas_v11’s own existing split, confirmed unchanged by anything here: the taxonomy, V_{derived} , location, timestamp, and expiry are all *defined* in Paper 1; link vectors are destined for Paper 2 (Assembly); the grouping mechanism that actually *uses* location and the time registers is destined for Inference. This entry adds specification to the first category. It does not change the other two.

10 Status

Mathematically complete and verified: the architectural decision, both encodings, the register structure, and the age/expiry split. Not mergeable into Paper 1 as a standalone edit, for the reason stated at the top of this file — the meta-vector taxonomy itself has to land first. One design question is left deliberately unresolved (expired-contribution filtering). Gap 2 is, for the first time since ideas_v1, fully specified rather than open; it is not yet closed in the paper, and should not be marked so until the taxonomy it depends on is merged.